Edgify Architecture

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Document Description

In this document, we would like to describe 2 of the most common Edgify architectures.
There are additional architectures that can be discussed if required (for example: the use of IP cameras / dealing with devices that are below the minimum requirements).
*** Please consult your Customer success Engineer that will assist customize the best solution that suits your environment and needs..**

Edgify Solution Components

Edgify Agent - The Device Agent is deployed on the Edge Devices.
It enables devices to execute real-time predictions as well as continuously collect samples for ongoing training.

Worker / Launcher - The Worker is in charge of the Training process.
In the Distributed architecture - the worker will be deployed on all edge devices and will be a part of a collaborative training architecture where all machines are taking part in the training process.
In Hybrid / Centralized architecture - the worker will be installed in 1 machine (can be an edgibox) which will store all images and that the training will be done on it.

Collaborator - The collaborative training module facilitates the collaborative training between the numerous edge devices. Consumes limited resources (can be installed on one of the edge devices).

- Advanced compression techniques
- Manage communication between the edge devices
- Auto recovery methods
- Installed in-store (on edge device or external dedicated device)

Dashboard - The dashboard enables Edgify's partners to manage the Edgify system and provide AI collaborative training capabilities on Edge devices.
High level capabilities:

- Manage Devices
- Aggregative view of the data collected on all the edge devices
- Run & Monitor training jobs
- Save trained Models
- Deploy and monitor Prediction Models

EdgiBox - An External device that can assist in environments that are not able to run training on the edge device directly.
Can be provided by Edgify or produced by the customer as long as it complies with the minimum requirements.

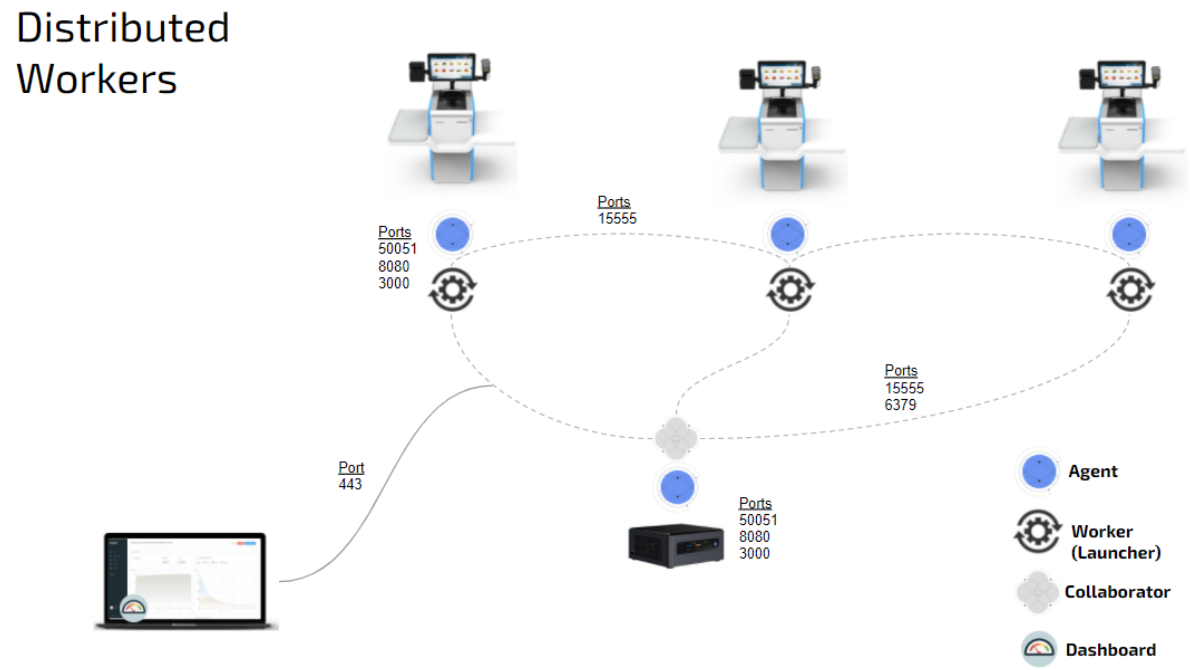
SetupUI and Data Collector - web application that allows testing the AI model, collecting data and deleting images from the db.

Distributed Architecture

Description

- Prediction and Training will be done using the existing edge device compute.
- An agent & worker is installed on each edge device
- The collaborator will be installed on an external device (low resources needed) - can be done on one of the edge devices.

Diagram



in-store collaborator -> cross-store collaborator (port 15555)

Minimum requirement

Edge Device:

Parameter	Min Spec	Comments
OS	Linux Ubuntu 16.04 (or above)	Doesn't have to be Ubuntu
Available Storage	10 GB	
Available CPU	Celeron Dual Core (or above)	We need about 50% of this cpu (one core of the dual cores)
Available Ram	2.5 GB	
Network internal LAN	100Mb	
Network external WAN	10Mb	

Collaborator device:

Parameter	Min Spec	Comments
OS	Linux Ubuntu 16.04 (or above)	Doesn't have to be Ubuntu
Available Storage	3 GB	
Available CPU	Celeron Dual Core (or above)	We need about 50% of this cpu (one core of the dual cores)
Available Ram	0.5 GB	
Network internal LAN	100Mb	
Network external WAN	10Mb	

Prerequisites

- **Camera 720p (or above), 20FPS (or above), 75°-96° FoV, Autofocus**
- **Linux OS**: Setup a Linux Machine with Ubuntu 16.04 or a higher version. (Note: If you are running on Windows, then we recommend using the VMWare Workstation Player as a Virtual Machine provider for running the Linux virtual guest machine). Native Windows installations will be provided upon request.
- **Docker Engine**: Install docker engine: Docker engine installation
- **Docker Compose**: Install docker compose: Docker compose installation
- **Docker Hub Account Access**: Edgify uses Docker Hub for hosting the containers repository. Ask Edgify for a username and password or ask for granting your own docker hub user access to the relevant containers repository
- **Device Tokens**: Coordinate with Edgify the amount and type of devices you plan to install - so that Edgify can supply you the relevant Tokens

Port Consideration

Edge Device

Port	Direction	Source/Destination	Notes
TCP 443	Outbound	GCP Edgify Portal	HTTPS - RestAPI + Websocket (Edgify Agent)
TCP 8008	*Outbound*	*On PREM Edgify Portal*	*HTTPS - RestAPI + Websocket (Edgify Agent)*
TCP 3000-3001 **TCP 8080**	Inbound	Maintenance machine	HTTP - For Setup-UI and Data Collector Access Consult Edgify SE on destination.
TCP 15555	Outbound	Cross site collaborator	Multi Site training

Edgify Portal (for On-Prem Portal)

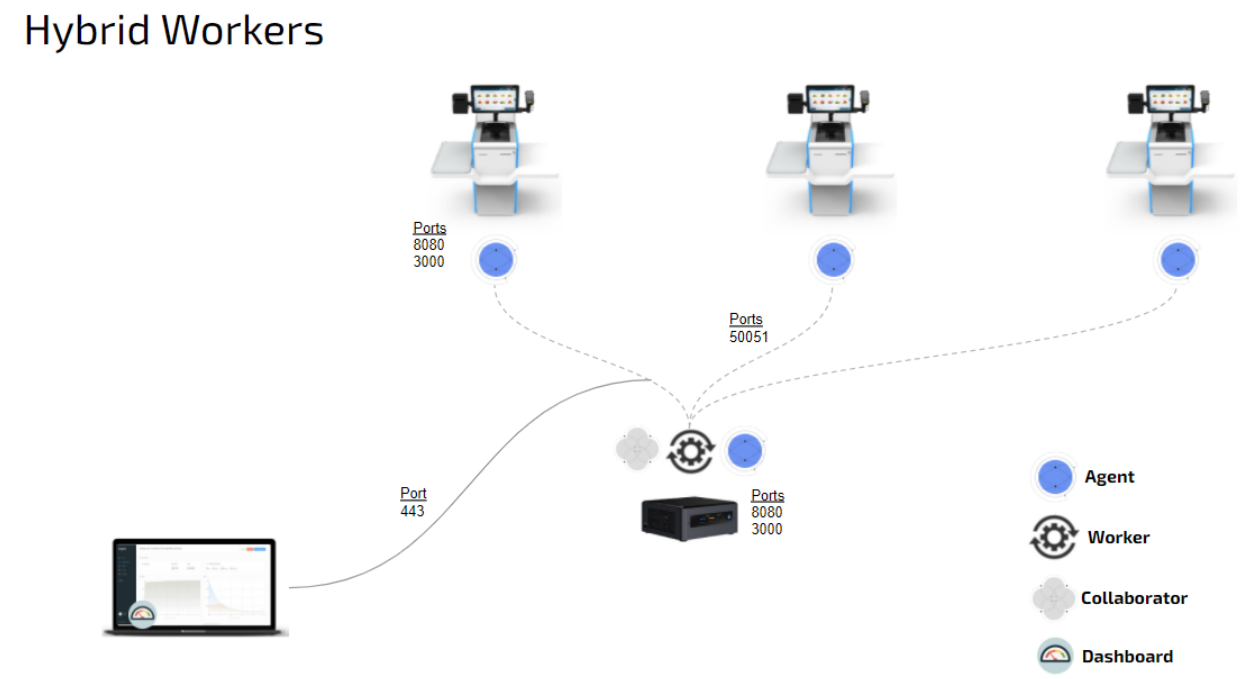
Port	Direction	Source/Destination	Notes
TCP 22	Inbound	Maintenance machine	SSH
TCP 8008	Inbound	Edge Device Maintenance machine	HTTP - RestAPI + Websocket (Edgify Agent) HTTP - Dashboard Access
TCP 443	Outbound	Docker Hub	HTTPS - For Pulling Docker Images from the Docker-Hub cloud

Hybrid Architecture

Description

- Prediction will be done using the agent that is installed on the edge device.
- Each agent will save the images on the “Edgibox” which will be used to run training.
- The worker & collaborator will be installed on the Edgibox.

Diagram



in-store collaborator -> cross-store collaborator (port 15555)

Minimum requirement

Edge Device:

Parameter	Min Spec	Comments
OS	Linux Ubuntu 16.04 (or above) / Windows 7 32 bit (or above)	
Available Storage	1 GB	
Available CPU	Celeron Dual Core (or above)	We need about 50% of this cpu (one core of the dual cores)
Available Ram	1 GB	
Network internal LAN	100 Mb	
Network external WAN	10 Mb	

Collaborator device (can be an EdgiBox):

Parameter	Min Spec	Comments
OS	Linux Ubuntu 16.04 (or above)	Doesn't have to be Ubuntu
Available Storage	50 GB	
Available CPU	Core-i5 or above	
Available Ram	8 GB	
Network internal LAN	100Mb	
Network external WAN	10Mb	

Prerequisites

- **Camera 720p (or above), 20FPS (or above), 75°-96° FoV, Autofocus**
- **Device Tokens**: Coordinate with Edgify the amount and type of devices you plan to install - so that Edgify can supply you the relevant Tokens

If using Linux Machine

- **Linux OS**: Setup a Linux Machine with Ubuntu 16.04 or a higher version. (Note: If you are running on Windows, then we recommend using the VMWare Workstation Player as a Virtual Machine provider for running the Linux virtual guest machine). Native Windows installations will be provided upon request.
- **Docker Engine**: Install docker engine: Docker engine installation
- **Docker Compose**: Install docker compose: Docker compose installation
- **Docker Hub Account Access**: Edgify uses Docker Hub for hosting the containers repository. Ask Edgify for a username and password or ask for granting your own docker hub user access to the relevant containers repository

Port Consideration

Edge Device

Port	Direction	Source/Destination	Notes
TCP 443	Outbound	GCP Edgify Portal	HTTPS - RestAPI + Websocket (Edgify Agent)
TCP 8008	*Outbound*	*On PREM Edgify Portal*	*HTTPS - RestAPI + Websocket (Edgify Agent)*
TCP 3000-3001 **TCP 8080**	Inbound	Maintenance machine	HTTP - For Setup-UI and Data Collector Access Consult Edgify SE on destination.
TCP 15555	Outbound	Cross site collaborator	Multi Site training

Edgibox

Port	Direction	Source/Destination	Notes
TCP 443	Outbound	GCP Edgify Portal	HTTPS - RestAPI + Websocket (Edgify Agent)
TCP 8008	*Outbound*	*On PREM Edgify Portal*	*HTTPS - RestAPI + Websocket (Edgify Agent)*
TCP 3000-3001 **TCP 8080**	Inbound	Maintenance machine	HTTP - For Setup-UI and Data Collector Access *Consult Edgify SE / CSE on destination.
TCP 50051	Inbound	PC-Scales POS Machines	GRPC - Saving Images from POS Machines and PC-Scales

Edgify Portal (for On-Prem Portal)

Port	Direction	Source/Destination	Notes
TCP 22	Inbound	Maintenance machine	SSH
TCP 8008	Inbound	Edge Device Maintenance machine	HTTP - RestAPI + Websocket (Edgify Agent) HTTP - Dashboard Access
TCP 443	Outbound	Docker Hub	HTTPS - For Pulling Docker Images from the Docker-Hub cloud